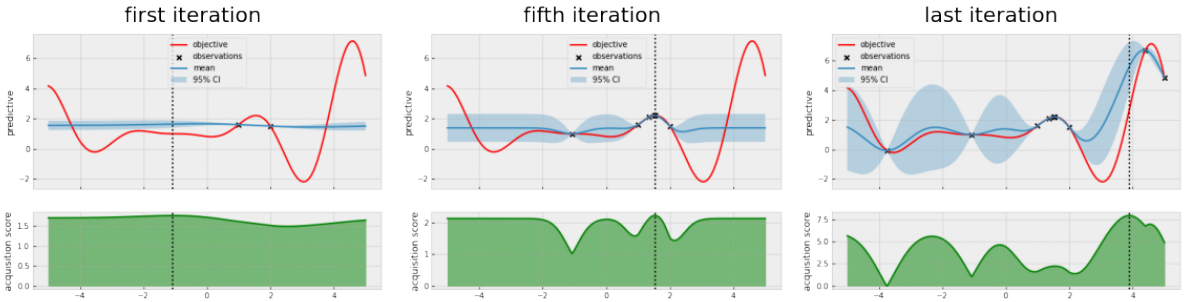


Figure A.10 Progress made by the adaptive version of the Upper Confidence Bound policy. The policy is able to escape the local optimum and get closer to the global optimum.

We see that the policy inspects a local optimum at the fifth iteration but ultimately is able to escape and get closer to the global optimum at the end.

A.5.2 Bayesian optimization for hyperparameter tuning

This exercise, implemented in `CH05/03 - Exercise 2.ipynb` applies BayesOpt to an objective function that simulates the accuracy surface of a support-vector machine model in a hyperparameter tuning task. There are multiple steps:

1. Recreate the Bayesian optimization loop in `CH04/03 - Exercise 2.ipynb`, including the outer loop that implements repeated experiments.
2. Run the Upper Confidence Bound (UCB) policy, setting the value of the tradeoff parameter at *1*, *3*, *10*, and *30*, and observe their aggregated performance.
   The value of the tradeoff parameter could be set when the policy object is initialized as below:

```
policy = botorch.acquisition.analytic.UpperConfidenceBound(
    model, beta=[some value]
)
```

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